



BEC Hamilton County Conversion Project – ERCOT Independent Review Study Status Update

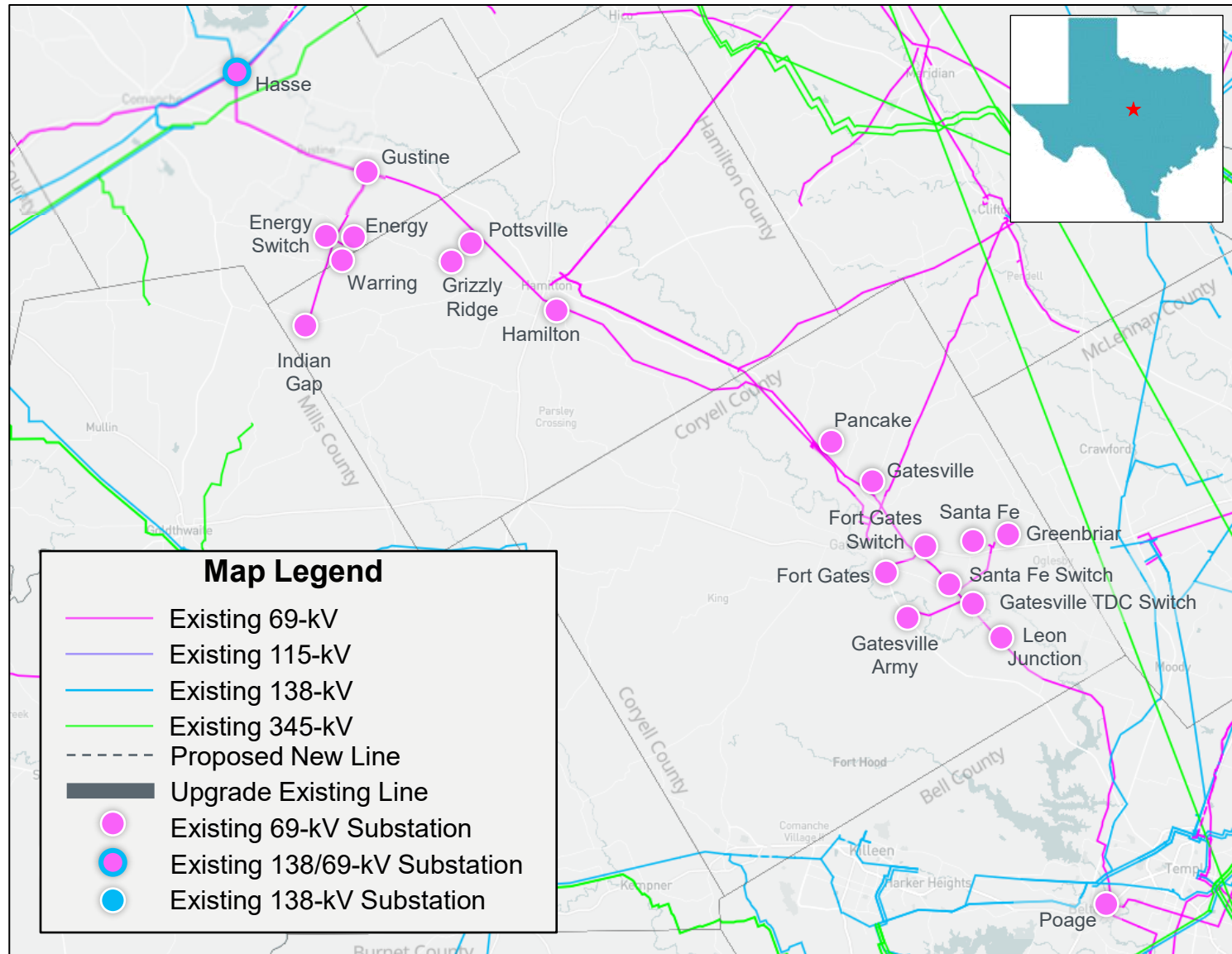
Travis Head

RPG Meeting
July 29, 2025

Introduction

- BEC submitted the Hamilton County Conversion Project for Regional Planning Group (RPG) review in February 2025
 - This Tier 2 project is estimated at \$90.0 million and will require a Convenience and Necessity (CCN)
 - Estimated in-service date (ISD) is Fall 2030
 - Is a GTC Exit Strategy for the Hamilton GTC
- BEC presented a project overview and ERCOT provided a project scope at the April 2025 RPG Meeting
 - <https://www.ercot.com/calendar/04292025-RPG-Meeting>
- ERCOT provided a project update June RPG Meeting
 - <https://www.ercot.com/calendar/06172025-RPG-Meeting>
- This project is currently under ERCOT Independent Review (EIR)

Study Area Map



Recap – Preliminary Results of Reliability Assessment – Base Case

Contingency Category*	Unsolved Power Flow	Voltage Violations	Thermal Overloads
P1	None	None	None
P3 (G-1+N-1)*	None	None	None
P6.2 (X-1+N-1)**	None	None	None
P7	None	None	None

*G-1 Generators tested: Grizzly Ridge Solar S1, Panda Temple C1, Logans Gap Wind W1

**X-1 Transformers tested: Hasse X1, Comanche Switch X1, Temple Switch X1

Preliminary Results of Planned Maintenance Outage Evaluation

- ERCOT conducted planned maintenance outage evaluation on the options
 - Load level in the North-Central Weather Zone was scaled down to 76.6% of its summer peak loads in the study base case, respectively based on ERCOT load forecast and historical load, in order to mimic the off- peak load condition
 - N-2 contingencies were tested as a proxy for N-1-1. Any applicable violating contingencies were further tested with system adjustments
 - The transmission elements in the Bell, Bosque, Comanche, Coryell, and Hamilton Counties were monitored in the maintenance outage evaluation
- Planned maintenance outage analysis results

Option	Voltage Violations	Thermal Overloads	Unsolved Power Flow
Base	None	None	None
1	None	None	None
2	None	None	None
3	None	None	None

Option 1 – BEC Proposed Project

- Convert existing Hasse to Gustine 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 418 MVA or greater, approximately 14.0-mile;
- Convert the existing Gustine, Energy, Energy Switch, Waring, Indian Gap, Pottsville, Hamilton, Gatesville, Fort Gates Switch, Fort Gates, Gatesville TDC Switch, Leon Junction, Poage 69-kV substations to 138-kV operation;
- Construct a new Gustine to Indian Gap 138-kV transmission line on double-circuit structures with normal and emergency ratings of 524 MVA or greater at 138-kV operation and 262 MVA or greater at 69-kV operation, approximately 11.65-mile;
- Rebuild existing Gustine to Energy Switch 69-kV transmission line as a 138-kV transmission line on double-circuit structures with normal and emergency ratings of 524 MVA or greater at 138-kV operation and 262 MVA or greater at 69-kV operation, approximately 5.1-mile;
- Rebuild existing Energy Switch to Energy 69-kV transmission line as a 138-kV transmission line on double-circuit structures with normal and emergency ratings of 524 MVA or greater at 138-kV operation and 262 MVA or greater at 69-kV operation, approximately 1.0-mile;

Option 1 – BEC Proposed Project cont'd

- Rebuild existing Energy Switch to Indian Gap 69-kV transmission line as a 138-kV transmission line on double-circuit structures with normal and emergency ratings of 524 MVA or greater at 138-kV operation and 262 MVA or greater at 69-kV operation, approximately 6.6-mile;
- Convert existing Gustine to Pottsville 69-kV transmission line to operate a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 11.0-mile;
- Convert existing Pottsville to Hamilton 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 7.6-mile;
- Convert existing Pottsville to Grizzly Ridge 69-kV transmission line to operate as a 138-kV transmission line with normal rating of 220 MVA and emergency rating of 242 MVA or greater, approximately 0.1-mile;
- Install two 138/69-kV autotransformers at the existing Pancake 69-kV substation with normal and emergency ratings of 60 MVA;
- Convert existing Hamilton to Pancake 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 21.9-mile;

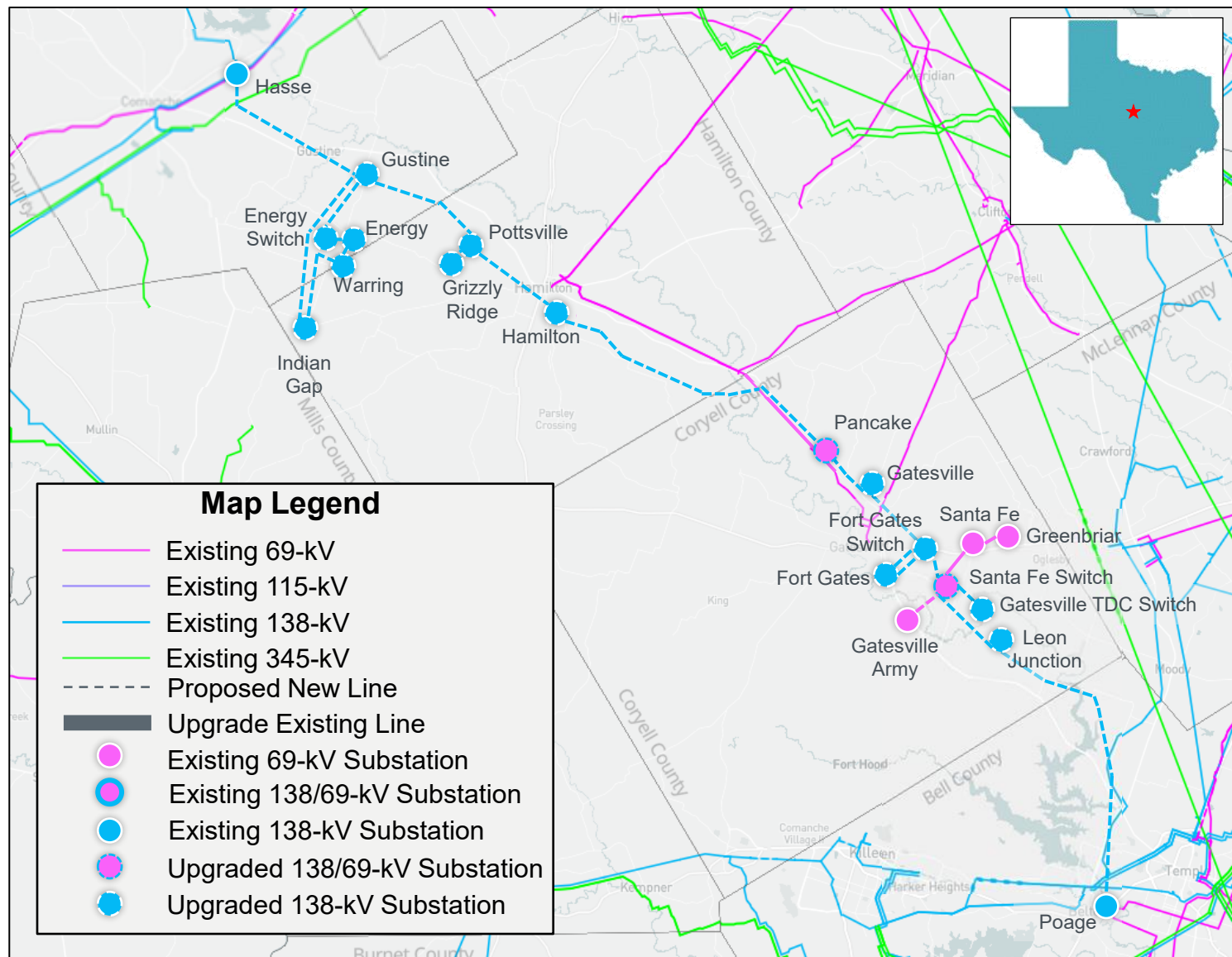
Option 1 – BEC Proposed Project cont'd

- Convert existing Pancake to Gatesville 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 4.0-mile;
- Convert existing Gatesville to Fort Gates 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 7.5-mile;
- Rebuild existing Fort Gates Switch to Fort Gates 69-kV transmission line as a 138-kV double-circuit transmission line with normal and emergency ratings of 524 MVA or greater at 138-kV operation and 262 MVA or greater at 69-kV operation, approximately 2.0-mile;
- Convert existing Fort Gates to Santa Fe Switch 69-kV transmission line as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 3.9-mile;
- Install two 138/69-kV autotransformers at the existing Santa Fe Switch 69-kV substation with normal and emergency ratings of 60 MVA;

Option 1 – BEC Proposed Project cont'd

- Rebuild existing Santa Fe Switch to Gatesville TDC Switch 69-kV transmission line as a 138-kV double-circuit transmission line with normal and emergency ratings of 524 MVA or greater at 138-kV operation and 262 MVA or greater at 69-kV operation, approximately 0.8-mile;
- Move existing Gatesville TDC Switch to Gatesville Army 69-kV tie line to connect to Santa Fe Switch creating the Santa Fe Switch to Gatesville Army 69-kV tie line;
- Convert existing Santa Fe Switch to Leon Junction 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 4.0-mile;
- Convert existing Leon Junction to Poage 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 24.4-mile; and
- Retire Poage autotransformer (Installed from Temple Area Improvements RPG).

Option 1 – BEC Proposed Project



Option 2 – ERCOT Proposed Option

- Convert the existing Gustine, Energy, Energy Switch, Waring, Indian Gap, Pottsville, and Hamilton 69-kV BEC owned substations to 138-kV operation;
- Remove existing 138/69-kV transformers at Hasse;
- Convert existing Hasse to Gustine 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 418 MVA or greater, approximately 14.0-mile;
- Convert existing Gustine to Energy Switch 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 5.1-mile;
- Convert existing Energy Switch to Energy 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 1.0-mile;
- Convert existing Energy Switch to Indian Gap 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 6.6-mile;

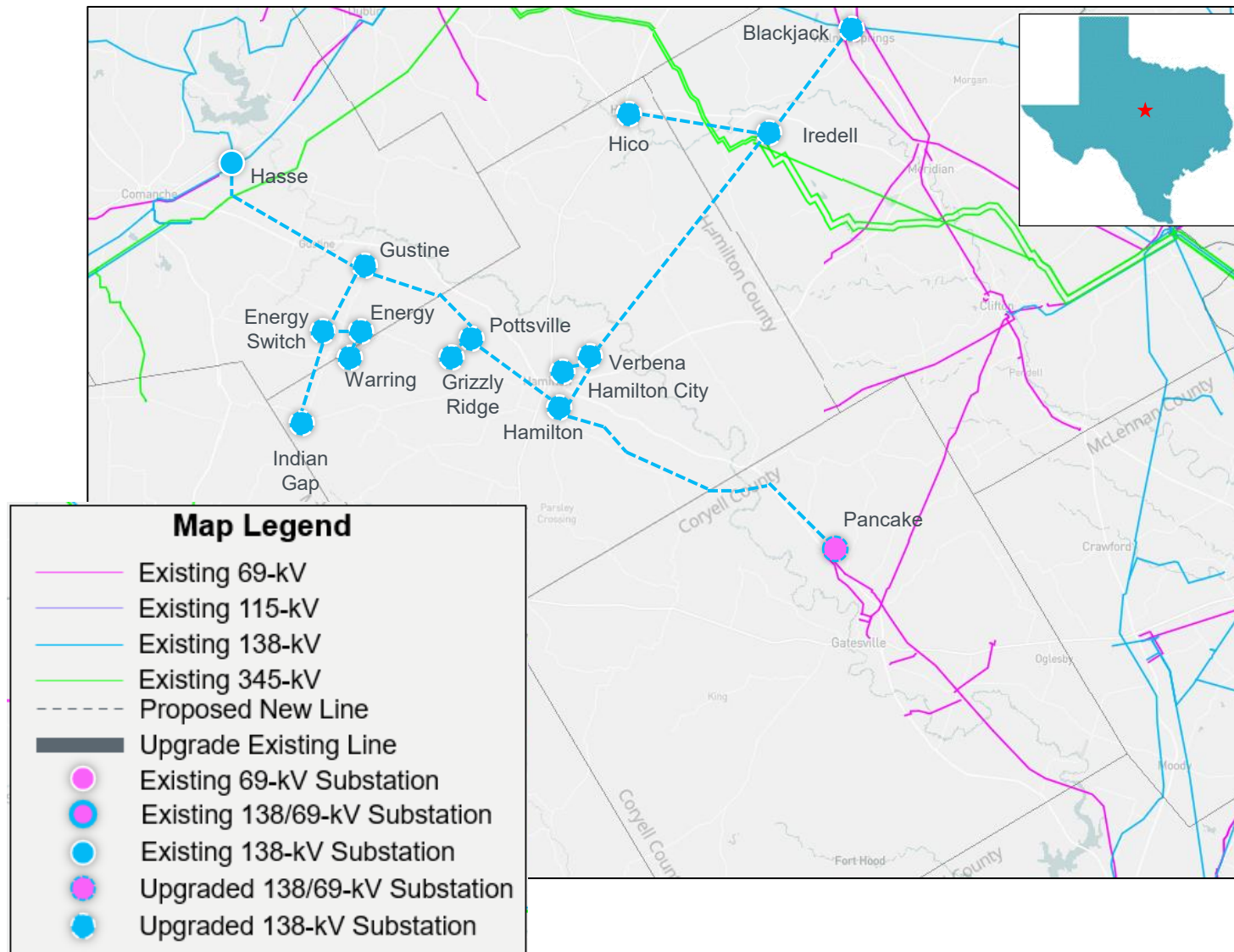
Option 2 – ERCOT Proposed Option cont'd

- Convert existing Gustine to Pottsville 69-kV transmission line to operate a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 11.0-mile;
- Convert existing Pottsville to Hamilton 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 7.6-mile;
- Convert existing Pottsville to Grizzly Ridge 69-kV transmission line to operate as a 138-kV transmission line with normal rating of 220 MVA and emergency rating of 242 MVA or greater, approximately 0.1-mile;
- Install two 138/69-kV autotransformers at the existing Pancake 69-kV substation with normal and emergency ratings of 60 MVA;
- Convert existing Hamilton to Pancake 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 21.9-mile;
- Convert the existing Hamilton City, Verbena, Hico, Iredell, and Blackjack 69-kV TNMP owned substations to 138-kV operation;

Option 2 – ERCOT Proposed Option cont'd

- Construct a new Hamilton to Verbena 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 2.0-mile;
- Convert existing Hamilton City to Verbena 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 0.9-mile;
- Convert existing Verbena to Iredell 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 22.3-mile;
- Convert existing Iredell to Blackjack 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 10.5-mile;
- Convert existing Iredell to Hico 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 9.9-mile; and
- De-energize the existing Verbena to Jonesboro 69-kV transmission line.

Option 2 – ERCOT Proposed Option



Option 3 – Combined Option

- Convert the existing Gustine, Energy, Energy Switch, Waring, Indian Gap, Pottsville, and Hamilton 69-kV BEC owned substations to 138-kV operation;
- Remove existing 138/69-kV transformers at Hasse;
- Convert existing Hasse to Gustine 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 418 MVA or greater, approximately 14.0-mile;
- Convert existing Gustine to Energy Switch 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 524 MVA or greater, approximately 5.1-mile;
- Convert existing Energy Switch to Energy 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 524 MVA or greater, approximately 1.0-mile;
- Convert existing Energy Switch to Indian Gap 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 524 MVA or greater, approximately 6.6-mile;

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- Convert existing Hamilton to Pancake 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 21.9-mile;
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Option 3 – Combined Option cont'd

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- Convert existing Iredell to Blackjack 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 10.5-mile;
- Convert existing Iredell to Hico 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 9.9-mile;
- De-energize the existing Verbena to Jonesboro 69-kV transmission line;

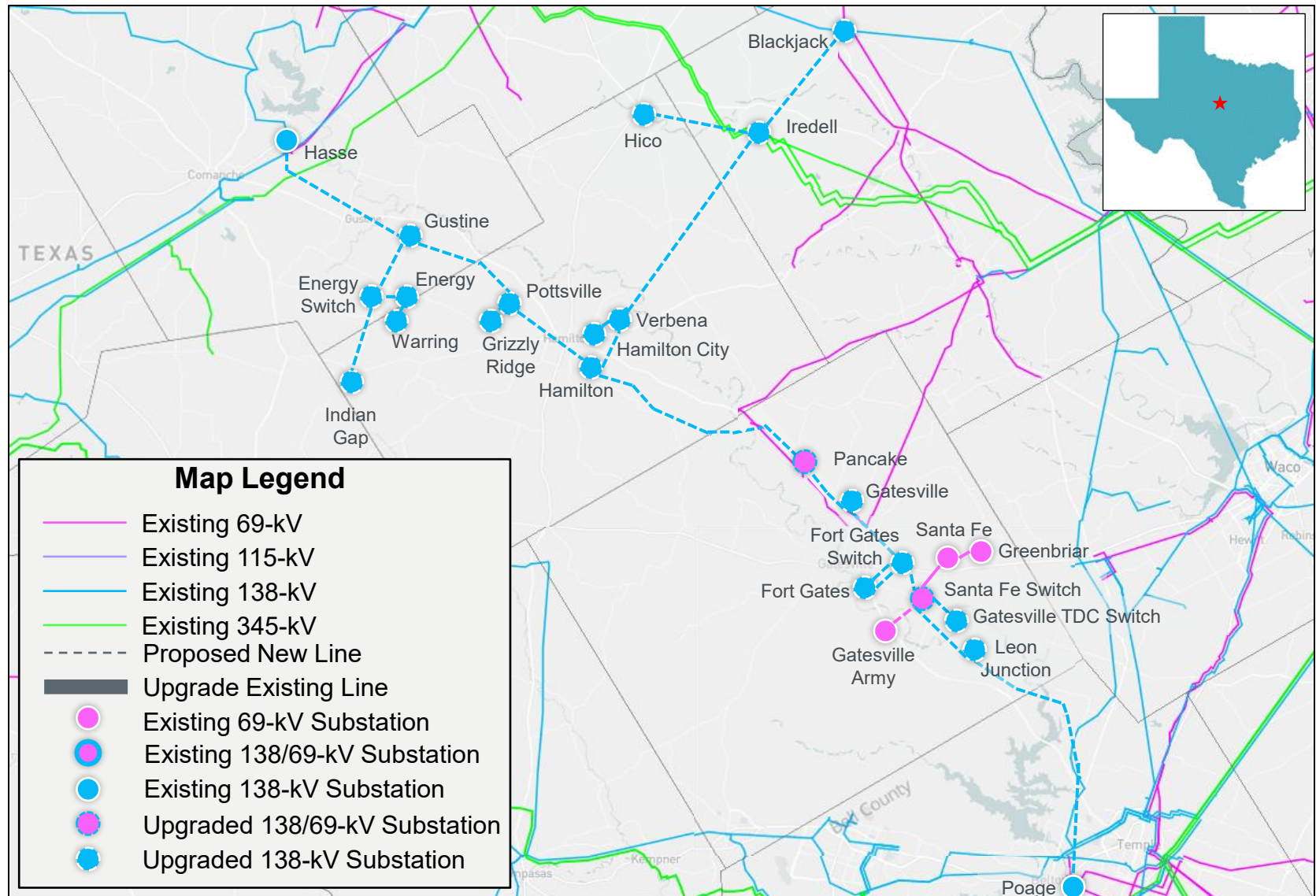
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- Rebuild existing Fort Gates Switch to Fort Gates 69-kV transmission line as a 138-kV double-circuit transmission line with normal and emergency ratings of 524 MVA or greater at 138-kV operation and 262 MVA or greater at 69-kV operation, approximately 2.0-mile;
- Convert existing Fort Gates to Santa Fe Switch 69-kV transmission line as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 3.9-mile;
- Install two 138/69-kV autotransformers at the existing Santa Fe Switch 69-kV with normal and emergency ratings of 60 MVA;

Option 3 – Combined Option cont'd

- Rebuild existing Santa Fe Switch to Gatesville TDC Switch 69-kV transmission line as a 138-kV double-circuit transmission line with normal and emergency ratings of 524 MVA or greater at 138-kV operation and 262 MVA or greater at 69-kV operation, approximately 0.8-mile;
- Move existing Gatesville TDC Switch to Gatesville Army 69-kV tie line to connect to Santa Fe Switch creating the Santa Fe Switch to Gatesville Army 69-kV tie line;
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- Convert existing Leon Junction to Poage 69-kV transmission line to operate as a 138-kV transmission line with normal and emergency ratings of 238 MVA or greater, approximately 24.4-mile; and
- Retire Poage autotransformer (Installed from Temple Area Improvements RPG).

Option 3 – Combined Option



Preliminary Results of Reliability Assessment – Options

	N-1		G-1+N-1*		X-1+N-1**	
Option	Thermal Violations	Voltage Violations	Thermal Violations	Voltage Violations	Thermal Violations	Voltage Violations
1	None	None	None	None	None	None
2	None	None	None	None	None	None
3	None	None	None	None	None	None

*G-1 Generators tested: Grizzly Ridge Solar S1, Panda Temple C1, Logans Gap Wind W1

**X-1 Transformers tested: Hasse X1, Comanche Switch X1, Temple Switch X1

Long-Term Load-Serving Capability Assessment

- Adjusted load up in substations in the Study Area
- Adjusted conforming load down outside of the North-Central Weather Zones to balance power
- Based on N-1 contingency limits

Option	Incremental Load-Serving Capability (~MW)
1	22.88
2	33.34
3	22.81

Preliminary Cost Estimate and Feasibility Assessment

- Transmission Service Providers (TSPs) performed feasibility assessments and provided preliminary cost estimates for the three options

Option	Cost Estimates (~\$M)	CCN Required (~Miles)	Feasibility
1	84.0	15.5	Yes
2	124.1	15.6	Yes
3	141.2	18.4	Yes

Updated – Economic Evaluation of Options

Option	Total Cost	Annual Cost Passing Requirement %	Consumer Revenue Passing Requirement %	Benefit to Cost Ratio (Annual Production Cost) %	Benefit to Cost Ratio (Consumer Revenue) %
O1 – BEC	84 \$M	13.0 %	12.7 %	0.3 %	18.1 %
O2 – ERCOT	124.1 \$M	13.0 %	12.7 %	0.8 %	12.9 %
O3 – Combined	141.2 \$M	13.0 %	12.7 %	2.2 %	10.3 %

Comparison of Options

	Option 1	Option 2	Option 3
Meets ERCOT and NERC Reliability Criteria	Yes	Yes	Yes
Meets TSP Criteria	Yes	No	No
Improves Long-Term Load-Serving Capability	Yes	Yes	Yes
Requires CCN (~miles)	Yes (15.5)	Yes (15.6)	Yes (17.5)
Project Feasibility	Yes	Yes	Yes
Cost Estimate* (~\$M)	84.0	124.1	141.2
Passes Consumer Revenue Passing Requirement	Yes	Yes	No

*Cost estimates were provided by the TSP(s)

ERCOT Preferred Option

- Option 1 is selected as the ERCOT preferred option because it:
 - Addresses the project need in the study area
 - Meets TSP Criteria
 - Improves Long-Term Load-Serving Capability
 - Requires the least amount of CCN mileage
 - Is the least expensive option
 - Passes the economic Consumer Revenue Passing Requirement

Additional Analyses

- Congestion Analysis
 - Congestion analysis and additional economic analysis will be performed for the preferred option using the 2024 RTP 2029 economic case
- Dynamic Analysis
 - Dynamic analysis will be performed to verify the preferred option solves the Hamilton GTC

Deliverables

- Tentative Timelines
 - Status updates at the future RPG meetings
 - EIR Report will be posted in the MIS in August

Thank you!



Stakeholder comments also welcomed through:

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